use. In the integrated system the data from different sensors is collected, processed, and displayed in a user-friendly way. Software provides effective command, control, and communication (C3). The systems are mostly scalable, sensor-agnostic, and user-friendly. Some examples of the existing systems are:

*Guardion Modular Counter–UAS System.* It provides a protective shield against the threat of unauthorised drones to both civil and military installations.The system is used by the German Armed Forces and airports to provide countermeasures against threats, such as hostile/illegal intrusion, smuggling, and terrorist attacks. The features include early warning, automatic detection and classification, powerful core intelligence and C3, and flexible customised deployment capability.

*DRDO-developed counter-drone system.* A counter-drone system has been developed by the DRDO to enable the Indian Armed Forces swiftly detect, intercept, and destroy small drones using both the soft kill and hard kill options. Soft kill refers to jamming the hostile drone, while hard kill neutralises the target drone.

The system uses radar that offers 360-degree coverage with detection of micro drones 4km away, electro-optical/infrared (EO/IR) sensors for detection of micro drones up to 2km, and a radio frequency (RF) detector to detect RF communication up to 3km. After detection it hands over the track for soft kill/hard kill.

The RF-jammer used for neutralisation is capable of jamming the signals of the hostile drone from a distance of 3km using the Global Navigation Satellite System (GNSS). The laser based hard kill system used can neutralise micro drones 150 metres to one kilometre away. The system is integrated through a command post.

*Anti-drone innovative systems.* Some innovative, low-cost and simple to use solutions are available for disabling the drones. SMASH 2000, an optical sight used in small arms for precision aiming, has been modified. The advanced version, SMASH 2000 Plus, uses built-in algorithms to provide capability to track and hit even very small drones flying at high speed at range up to 120 metres, both during day and nighttime.

*SMASH 2000, an optical sight used in small arms for precision aiming at drones (Source: https://www.jpost.com)*

DroneKiller is another standalone handheld anti-drone device available from IXI EW. The device using software-defined radio technology, operating on seven frequency bands, can neutralise drones up to a range of 1000 metres and be in active mode for up to two hours (eight hours in standby), weighs 3.4kg and is comfortable to carry.

## Conclusion

Drones are here to stay and we have to face them. They pose ever-increasing, multifaceted threat to civil and military infrastructure, assets and people and are being increasingly used for nefarious activities like terror and illegal surveillance, as improvised explosive devices (IEDs), and to transport arms, drugs, and other contraband. Drones have been adopted by corporates and civil bodies for urban mobility, delivery services, crisis management, disaster recovery, and emergency supply, to name a few.

From the military point of view, we are now entering the 'second drone age' in which countries are increasingly employing drones for covert operations and tactical use. In bigger role, the counter-drone systems work in tandem with reconnaissance, intelligence, surveillance and target acquisition (RISTA), and electronic warfare systems to tackle the threat of stand-alone drones as well as swarms.

The advent of the Internet of Things (IoT) and 5G technologies is opening up new possibilities to operate drones. The 4th generation GPS systems are becoming increasingly more interference-proof. Long Term Evolution (LTE)—a high-performance interface for cellular mobile communication systems to increase the capacity and speed of networks—is enabling drones to operate at theoretically unlimited ranges without RF links.

With a drone flying at 20 metres per second, taking less than a minute to cover one kilometre, an operator will have only a few seconds to react from the time of detection and determining the drone's intention. In such a dynamic situation, the state-of-the-art counter systems need to be developed and continuously updated.

The good news is that the counter-drone technologies are becoming smarter, more sophisticated, and are using innovative approaches, such as machine learning, sensor fusion, cognitive and holographic radars, and augmented reality. Nevertheless, experts agree that no single 'silver bullet' will ever exist to cope with the drone threat.

Finally, innovative and out-of-the-box thinking is essential to face the problems modern warfare brings, such as uncontrolled and fast-developing technologies, unestablished legal frameworks, and complex ethical questions. EFY